

Samarth Aradhya

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EDUCATION

Purdue University

Bachelor of Science in Cybersecurity, Planned double major in Mechanical Engineering

West Lafayette, IN

Aug. 2025 – May 2029

EXPERIENCE

Code Ninjas Tutor

San Ramon Code Ninjas

May 2022 – March 2025

San Ramon, CA

- Taught students foundational programming in JavaScript, Python, and Robotics, guiding students through debugging and project development.
- Initiated and led the development of a new 3D Printing and Modeling Camp introducing students to CAD software and physical development of their ideas.

Head Structural Designer

California High School Robotics Club

Sep. 2022 – Mar. 2024

San Ramon, CA

- Served as head structural designer for my team and led design of drivetrains and intake using OnShape CAD software..
- Collaborated in a 4-person team to iterate chassis prototypes under strict weight, torque, and sizing constraints.

Eagle Scout

Scouts of America, Boy Scouts Troop 84

March 2019 – Present

San Ramon, CA

- Earned Eagle Scout rank; directed 20+ volunteers across 100+ community service and conservation build hours.
- Planned multi-stage troop events and logistics, including equipment transport and route mapping during dynamic conditions.

PROJECTS

Quadruped Locomotion Robot | C++, ESP32, Kinematics, PWM, Bluetooth

Jan. 2026 – Present

- Built an 8-DOF legged robot to evaluate multi-axis joint actuation, weight distribution, and gait stability.
- Implemented custom C++ kinematics firmware to synchronize multi-servo PWM signals for walking and turning cycles.
- Calibrated joint offsets and dynamic timing loops to mitigate physical backlash and maintain balance during movement.

Finger Tracking Glove | ESP32, Flex Sensors, I2C/SPI

September 2025 – Present

- Developed a wearable glove that tracks finger flexion using calibrated flex sensors and streams data real time using an ESP32 micro controller.
- Designed a digital 3D model that accurately shows hand position based on information from glove.
- Working on integration with a robotic hand for remote manipulation.

ESP32 Wi-Fi Robotic Spider | Embedded C++, WebSockets, CAD

October 2024 – December 2024

- Built a mobile robotic platform with real time video streaming using an ESP32-CAM chip.
- Implemented a real-time control system built on a Web Application.
- Worked with low level debugging and multi-system synchronization for field robotics applications.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, Lua

Hardware & Tools: KiCad, SolidWorks, Tinkercad, 3D Slicers, Oscilloscope, Multimeter, SMD Soldering, Git, Linux

Embedded & Protocols: ESP32, RP2040, Arduino, I2C, SPI, UART, PWM, ADC

ACTIVITIES

Purdue University

Formula Baja, Purdue Badminton Club

West Lafayette, IN

Fall 2025 – Present